Progress Note: 2-Period Igami Model with Agglomeration

14.273 Mini Project

April 14, 2026

1 Overview

This note documents the first extension of Igami (2017) to incorporate **innovation spillovers / agglomeration effects** in a simplified 2-period setting. The goal is to understand whether the presence of other innovative firms raises or lowers the incentive to adopt new technology.

2 Model

2.1 State and Firm Types

The market state at each period t is $s_t = (n_o, n_b, n_n, n_{pe})$, where:

- n_o: old incumbents (have *not* adopted new technology),
- n_b: "both" incumbents (have adopted; produce with new tech),
- n_n: new entrants (using new technology),
- n_{pe}: potential entrants.

Active firms in the product market are those with $n_o + n_b + n_n > 0$. "Both" firms are multi-product: they continue to sell old-gen (at cost c_o) while also selling new-gen (at cost $c_{n,t}$). This is the source of the **cannibalization effect**: by expanding new-gen supply, a "both" firm depresses the old-gen price it receives, creating a disincentive to innovate that is absent for pure new entrants.

2.2 Timing Within Each Period

1. **Product market:** Cournot competition in *two differentiated markets* (old-generation and new-generation products) with linear inverse demand:

$$P_o = A - \frac{B}{M} Q_o - \frac{\rho B}{M} Q_n, \tag{1}$$

$$P_n = A - \frac{B}{M} Q_n - \frac{\rho B}{M} Q_o, \tag{2}$$

where $\rho \in [0, 1)$ is the cross-market substitution parameter. Old firms (n_o) compete only in the old-gen market at cost c_o. New entrants (n_n) compete only in the new-gen market at cost $c_{n,t}$. "Both" firms (n_b) are *multi-product*: they sell in both markets simultaneously and internalize the **cannibalization** effect — their new-gen output reduces demand for their own old-gen product.

2. **Private shocks:** Each firm draws i.i.d. cost shocks $\varepsilon_{i,a} \sim \text{Gumbel}(0, \sigma)$ for each action a.

3. **Discrete choices** (sequential order: old → both → new → pe):

   - old: $\{\text{stay}, \text{innovate}, \text{exit}\}$
   - both/new: $\{\text{stay}, \text{exit}\}$
   - pe: $\{\text{enter}, \text{quit}\}$

   Each type observes the *realized* outcomes of all earlier movers before making its own decision.

4. **Transition** to period $t + 1$.

2.3 Agglomeration Extension

The new-technology marginal cost depends on the number of firms already using it:

$$c_{n,t} = c_{n,0} - \gamma \cdot (n_{b,t} + n_{n,t}), \qquad \gamma \geq 0. \tag{3}$$

When $\gamma = 0$ the model reduces to the Igami baseline. When $\gamma > 0$, a larger innovative cluster lowers costs for all new-tech firms (knowledge spillover / agglomeration effect).

2.4 Value Functions and CCPs

Because firms move sequentially within each period and each type observes the *realized* outcomes of all earlier movers, each type's decision problem is a genuine **single-agent problem**: earlier-mover uncertainty has been resolved, and later-mover choices enter only through pre-computed CCPs. Combined with i.i.d. EVT1 private cost shocks, this yields closed-form logit CCPs at each stage:

$$P(a \mid \tau, \tilde{s}) = \frac{\exp\big(\bar{V}_a(\tau, \tilde{s})/\sigma\big)}{\sum_{a'} \exp\big(\bar{V}_{a'}(\tau, \tilde{s})/\sigma\big)}, \tag{4}$$

where $\tilde{s}$ is the *realized intermediate state* after all earlier movers have acted, and $\bar{V}_a$ is the systematic utility:

$$\bar{V}_a(\tau, \tilde{s}) = \pi_\tau(s_0) + \text{cost}(a) + \beta\, \mathbb{E}[V_{t+1}(\tau', s') \mid \tilde{s},\, a_i = a,\, \text{CCP}_{\text{later}}]\,. \tag{5}$$

Here $\text{CCP}_{\text{later}}$ denotes pre-computed CCPs of same-type peers and all later-moving types; flow profits $\pi_\tau(s_0)$ are evaluated at the period-opening state.

2.5 Solution Method

The model is solved by **one-pass backward induction**, following Igami (2017):

1. **Terminal period ($t = 1$):** $V_1(\tau, s_1) = \pi_\tau(s_1)$, where Cournot profits use the agglomeration-adjusted cost $c_{n,1} = c_{n,0} - \gamma(n_{b,1} + n_{n,1})$.

2. **Period $t = 0$:** For each state s_0, stages are solved in *reverse move order* (PE $\rightarrow$ new $\rightarrow$ both $\rightarrow$ old), exploiting the sequential structure:

   - **Stage 4 (PE):** Each PE firm observes the full realized intermediate state $\tilde{s}$ and solves a binary logit $\{\text{enter}, \text{quit}\}$. CCPs are pre-computed for all intermediate states reachable from s_0.
   - **Stage 3 (new):** Each new firm solves a binary logit $\{\text{stay}, \text{exit}\}$, conditioning on pre-computed PE CCPs to integrate over the PE-stage continuation value.
   - **Stage 2 (both):** Each both firm solves a binary logit, conditioning on new + PE CCPs. Innovators locked in from Stage 1 (k_{io} old firms that chose innovate) are added to n_b before this stage.
   - **Stage 1 (old):** Each old firm solves a multinomial logit $\{\text{stay}, \text{innovate}, \text{exit}\}$, conditioning on all lower-stage CCPs.

   No fixed-point iteration is needed within the period: the sequential structure makes each stage a closed-form single-agent problem given pre-computed downstream CCPs. Marginal CCPs for later-moving types are recovered by integrating over the equilibrium transition distribution (products of independent Binomial/Multinomial draws).

3 Calibration

The calibration is illustrative (not estimated from data) and is summarised in Table 1.

Table 1: Illustrative calibration

Parameter	Description	Value
A	Demand intercept	3.0
B	Demand slope	1.0
M	Market size	1.0
c_o	Old-tech marginal cost	1.5
$c_{n,0}$	New-tech baseline cost	0.5
β	Discount factor	0.9
κ	Innovation cost	0.3
ϕ	Entry cost	0.2
σ	EVT1 scale	1.0
γ	Agglomeration (comparative statics)	varied
ρ	Cross-market substitution	0.5

The initial state of interest is $s_0 = (n_o, n_b, n_n, n_{pe}) = (4, 1, 1, 2)$, loosely inspired by the early HDD market structure in Igami (2017).

4 Results

4.1 Baseline vs. Agglomeration

Table 2 compares equilibrium CCPs at s_0 for $\gamma = 0$ (baseline) and $\gamma = 0.05$.

Table 2: CCPs at $s_0 = (4, 1, 1, 2)$: baseline vs. agglomeration ($\gamma = 0.05$)

	$\gamma = 0$	$\gamma = 0.05$
$P(\text{innovate} \mid \text{old})$	0.3135	0.3206
$P(\text{enter} \mid \text{pe})$	0.5061	0.5139

Agglomeration raises both the innovation and entry rates. Intuitively, a larger innovative cluster lowers future costs for new-tech firms, increasing the expected continuation value of innovating or entering.

4.2 Comparative Statics in γ

Table 3 reports equilibrium CCPs as γ varies from 0 to 0.30, and Figure 1 plots the same.

Table 3: CCPs vs. agglomeration parameter γ

γ	$P(\text{innovate}\|\text{old})$	$P(\text{enter}\|\text{pe})$
0.00	0.3135	0.5061
0.05	0.3206	0.5139
0.10	0.3282	0.5221
0.15	0.3314	0.5258
0.20	0.3318	0.5263
0.25	0.3315	0.5260
0.30	0.3316	0.5260

4.3 Comparative Statics in ρ (Cannibalization)

Figure 2 plots equilibrium CCPs as the substitution parameter ρ varies from 0 (independent markets) to 0.95 (near-homogeneous goods), holding $\gamma = 0.05$.

4.4 Discussion

Three features of the comparative statics are notable:

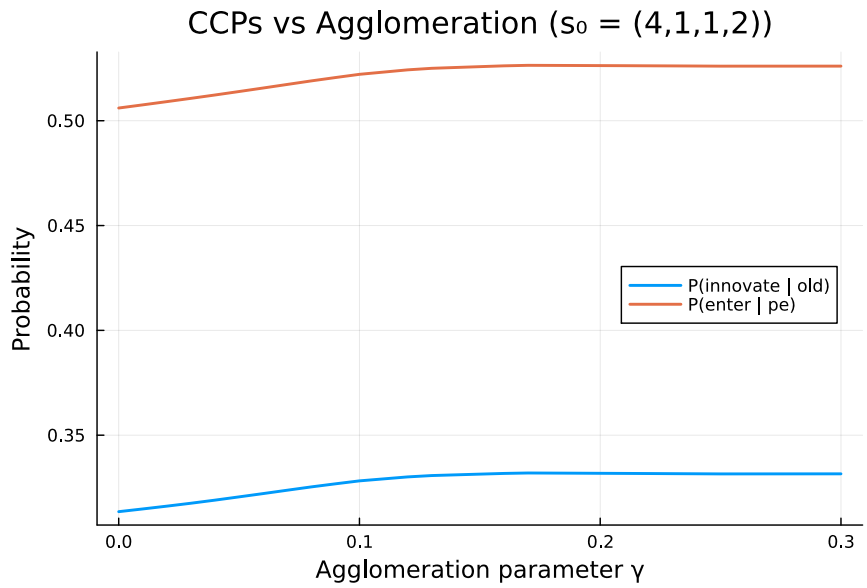

Figure 1: $P(\text{innovate} \mid \text{old})$ and $P(\text{enter} \mid \text{pe})$ as functions of γ at $s_0 = (4, 1, 1, 2)$.

1. **Positive agglomeration effect:** Higher γ unambiguously raises the innovation rate for small values of γ. This confirms the model's central prediction: knowledge spillovers encourage adoption.

2. **Cannibalization deters innovation:** Higher ρ (stronger substitution between old-gen and new-gen) reduces the profit of "both" firms by depressing old-gen prices. In the calibration, per-firm "both" profit falls from ≈ 0.531 at $\rho = 0$ to ≈ 0.327 at $\rho = 0.5$. This is the central mechanism in Igami (2017): incumbents are reluctant to innovate because doing so cannibalizes their existing old-gen revenue.

3. **Saturation:** The innovation rate saturates near 1/3 for large γ. This is a feature of the EVT1 / logit structure: with large σ relative to payoff differences, all actions approach equal probability. In a richer calibration (smaller σ), the effect would be more pronounced.

5 Regional Extension and Counterfactuals

5.1 Regional Structure

To study the interaction between industrial policy and agglomeration we extend the state space with a **region** attribute. There are $R = 3$ regions (labelled $r \in \{1, 2, 3\}$); each firm is born into a region and never leaves it, and potential entrants sit in per-region pools $n_{pe,r}$. Product-market competition remains **global** — all active firms meet in the same two-market Cournot game of equations (1)–(2) — but agglomeration spillovers are now **local**:

$$c_{n,r,t} = c_{n,0} - \gamma_r \cdot \big(n_{b,r,t} + n_{n,r,t}\big), \qquad r = 1, \ldots, R. \tag{6}$$

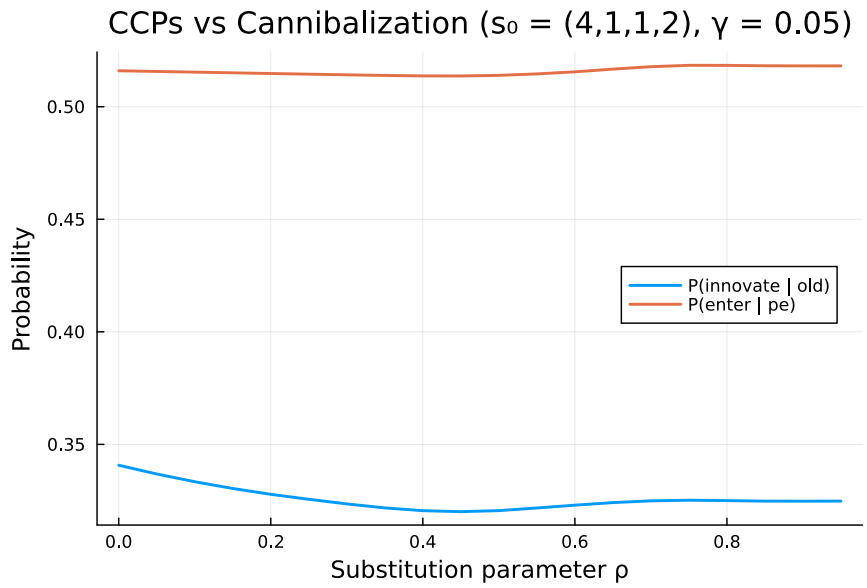

Figure 2: $P(\text{innovate} \mid \text{old})$ and $P(\text{enter} \mid \text{pe})$ as functions of ρ at $s_0 = (4, 1, 1, 2)$, $\gamma = 0.05$.

Because $c_{n,r,t}$ varies across regions, firms of type "both" and "new" face *heterogeneous* marginal costs in the new-gen market. The Cournot stage therefore loses symmetry across regions, and equilibrium quantities are obtained by solving an asymmetric FOC system of up to $4R$ equations per state. Within-region slots remain symmetric (all firms in the same $(type, r)$ cell pick the same quantity), so per-firm profits are still characterised by $\pi_{o,r} = bq_{oo,r}^2$, $\pi_{n,r} = bq_{nn,r}^2$, and $\pi_{b,r} = b(q_{bo,r}^2 + q_{bn,r}^2) + 2s\, q_{bo,r} q_{bn,r}$ where $b = B/M$ and $s = \rho B/M$.

Each firm privately captures the lower cost when it innovates but does not internalize the benefit it confers on regional peers — a standard **local agglomeration externality** that creates a wedge between the social and private return to innovation and opens the door to regionally targeted industrial policy.

5.2 Solution with Regional States

The sequential-move structure (old $\to$ both $\to$ new $\to$ pe) carries over unchanged: within each stage firms in the same $(type, r)$ slot still solve a single-agent problem, and i.i.d. EVT1 shocks still deliver closed-form logit CCPs. The only change is that each stage now has *one CCP per region*, and the three regional CCPs are jointly pinned down by a Jacobi fixed point on the stage's logit system.

Concretely, the solver uses layered caches so that the stage-4 (PE) problem — which depends only on the post-new-stage state — is solved once and reused across initial states via $\text{EV}_{\text{after pe}}[s_{an}]$. Stage 3 (new) cascades through that cache, stage 2 (both) through $\text{EV}_{\text{after new}}$, and stage 1 (old) through $\text{EV}_{\text{after both}}$. Because the state space now has $4R = 12$ integer coordinates, we cap $N_{\max} = 6$ on this branch (about 1.8×10^4 enumerated states) and pick initial states with total firm count ≤ 6.

5.3 Baseline and Counterfactuals

We solve the regional model at two initial states, both with total firm count 6:

- **Symmetric s_0:** $s_0^{\text{sym}} = \big((1,1,1),(1,1,1),(0,0,0),(0,0,0)\big)$ — one old incumbent and one "both" incumbent in each region, no entrants.
- **Clustered s_0:** $s_0^{\text{clust}} = \big((1,1,1),(3,0,0),(0,0,0),(0,0,0)\big)$ — same number of firms of each type, but all three "both" incumbents are concentrated in region 1.

Starting from s_0^{sym} with uniform $\gamma_r = 0.05$ we verify a **symmetry check**: the three regions' CCPs are identical to machine precision, consistent with the theoretical invariance of the model under region permutations when the state and parameters are symmetric.

Two counterfactuals are then run against this baseline. **Counterfactual A** subsidises innovation in region 1 by tripling its agglomeration strength to $\gamma_1 = 0.15$ while leaving $\gamma_2 = \gamma_3 = 0.05$. **Counterfactual B** holds γ_r uniform but starts from the clustered initial state s_0^{clust}, asking whether a historical concentration of innovators is enough to lock in innovation leadership even without a policy tilt.

Table 4: Regional innovation CCPs $P(\text{innovate} \mid \text{old}, r)$ at each scenario. Baseline uses s_0^{sym} with uniform $\gamma = 0.05$; CF-A sets $\gamma_1 = 0.15$; CF-B uses s_0^{clust}.

Scenario	$P(\text{innov}\|o, r{=}1)$	$P(\text{innov}\|o, r{=}2)$	$P(\text{innov}\|o, r{=}3)$
Baseline	0.3331	0.3331	0.3331
CF-A: subsidy r1	0.3590	0.3265	0.3265
CF-B: cluster r1	0.3366	0.3233	0.3233

5.4 Discussion

Three findings stand out in Table 4 and Figure 3:

1. **Targeted subsidies work, but leak.** Raising γ_1 from 0.05 to 0.15 moves region 1's innovation probability from 0.3331 to 0.359, a 0.3331-to-0.359 gain of roughly two and a half percentage points. Regions 2 and 3 fall from 0.3331 to 0.3265— a small but clearly negative spillover driven entirely by the *global* product market: more innovation in region 1 lowers expected new-gen prices everywhere, eroding the option value of innovating in the other regions.
2. **Historical clusters alone are weak.** Starting from s_0^{clust} the incumbent cluster in region 1 barely moves innovation there (0.3366 vs. 0.3331 in the symmetric baseline): with only one remaining old firm in region 1, the region's marginal old firm is already close to saturating its own local spillover. The larger effect shows up in regions 2 and 3, whose CCPs fall to 0.3233 because of the global-market pressure from the concentrated region-1 innovators. A one-shot historical endowment is therefore a much weaker policy lever than a continuing subsidy.

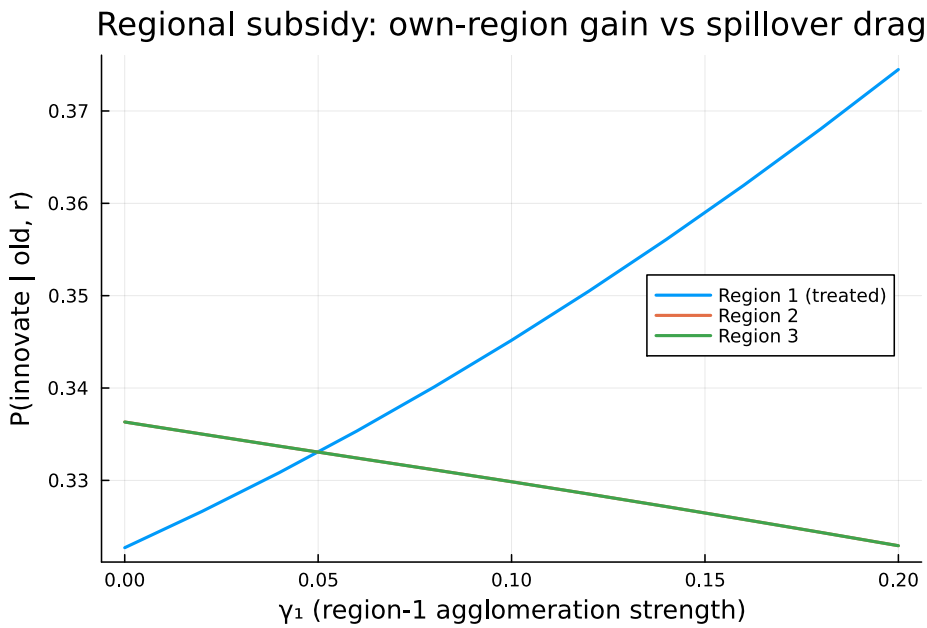

Figure 3: Regional innovation rates as γ_1 varies while $\gamma_2 = \gamma_3 = 0.05$ is held fixed. Region 1 responds positively to its own subsidy; regions 2 and 3 are slightly depressed through the global Cournot channel.

3. **The spillover wedge is still local.** Comparing CF-A and CF-B makes clear that what drives region 1's innovation is not how many innovators are in the region today but the *marginal reduction in new-tech cost* each additional innovator delivers. The policy CF-A tilts that reduction directly, which is why it dominates a pure history-based CF-B at the same total cluster size.

The two experiments illustrate how the regional model bridges Igami's replacement-effect story with the industrial-policy literature: even a modest, regionally targeted spillover subsidy reshapes the geographic distribution of innovation at the cost of a small drag on the rest of the world, with magnitudes that depend on the level of cannibalization ρ and the strength of the local externality γ_r.

6 Next Steps

1. Replace the illustrative calibration with Igami's (2017) structural estimates (Table 3 of the paper).
2. Extend to $T > 2$ periods (the HDD panel spans 1977–1999).
3. Richer counterfactuals on the regional branch: bilateral spillover pools (e.g. regions 1–2 share an enlarged effective N^{inn}), endogenous region choice at entry, and a welfare accounting that nets consumer surplus against the fiscal cost of the subsidy.
4. Study welfare and industrial policy implications (e.g., R&D subsidies).